

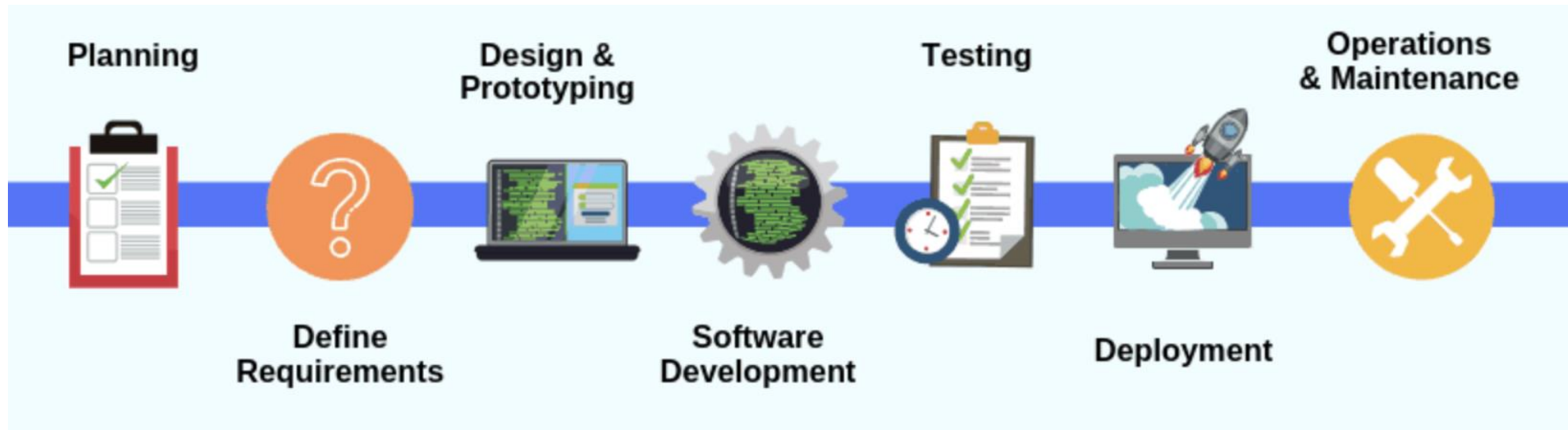
Tools for software development

Choosing a testing tool



Testing implementation

- testing is part of the software life cycle
- statistics:
 - 29% of testing budget (2021)



Testing and automation

- testing requires the presence of an employee
 - at design and launch
 - suitable for smaller projects
- automated testing
 - proposed by an employee
 - better integration into development
- the decision depends on the type of project and the team's decisions



“ 3P ” on software testing and automation approach

- **Product / Platform**

- type of tool to test before starting
- as long as the tool fits
 - brings benefits
 - increases testing efficiency

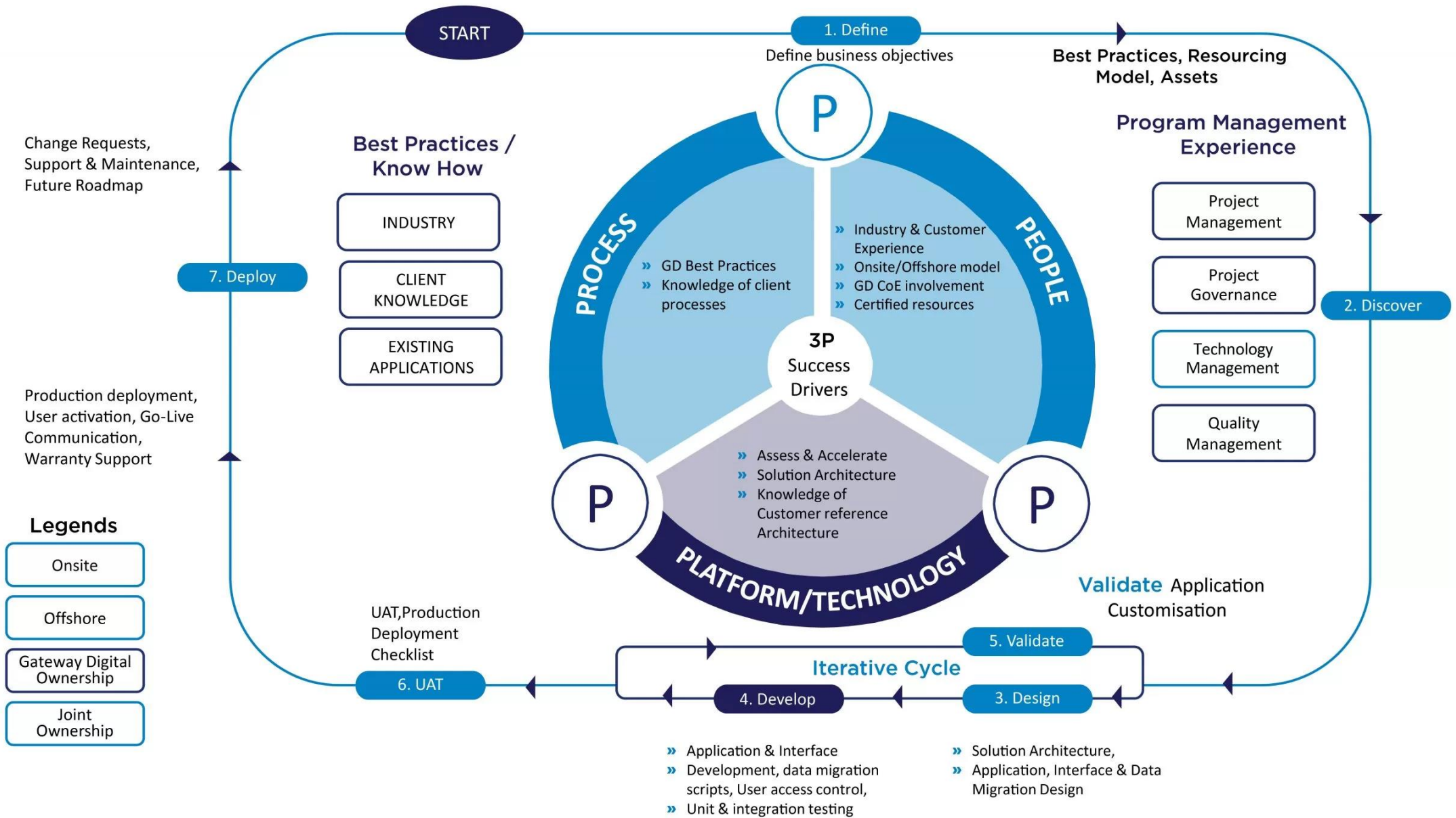
- **People**

- the team must accept the decision
- ability to adapt to changes

- **Processes**

- method of testing
- in the selected tool with the team design





Benefits of using software tools

- **higher test coverage**
 - with automation we can run tests more often
- **saving money and time**
 - when running tests
- **support for multiple platform types**
 - verification for all types
- **bug-free software versions**
 - or at least with fewer numbers than if we couldn't test
- **faster finding and repairing defects**
 - continuous and early search and repair
- **faster release of newer versions**
 - we can deliver newer versions faster and theoretically without errors

Characteristics of a *good* testing tool

- easy and simple to use
- debugging support
- robust object identification
- support for tests in the environment
- database testing
- cross-platform support



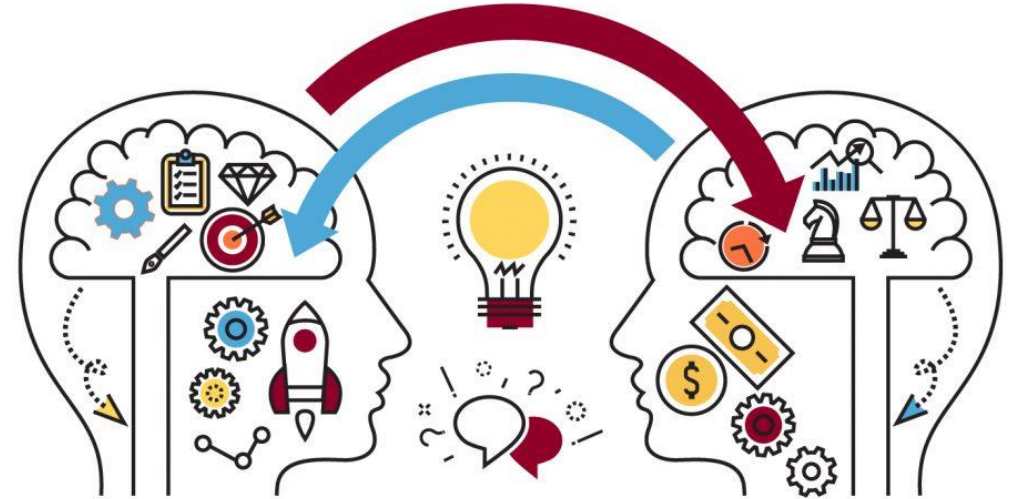
How to choose a tool?

- understanding the project and requirements
- evaluation of currently used tools
- evaluation of selected instruments according to criteria
- creating a decision table



Understanding the project and requirements

- project type
 - web, mobile, desktop
- project scope
 - total utilization
- technical expertise
 - development experience
- need for automation
 - the resulting evaluation of whether it will be beneficial to the project



Evaluation of currently used tools

- Without comparison we won't know which is better.
- evaluation of the pros and cons of a given tool
- creating an experiment with multiple tools
- resulting evaluation of e.g. time for implementation, running and administration



Evaluation of selected instruments according to criteria

- example of criterion selection
 - easy to use
 - OS support for development and testing
 - supported platform
 - support for multiple languages
 - budget and price of the tool
 - possibility of creating reports



Creating a decision table

- evaluation of criteria for all instruments
- currently used tool
 - "benchmark" for comparing new tools
- we can get information
 - by real testing
 - by googling
 - reviews



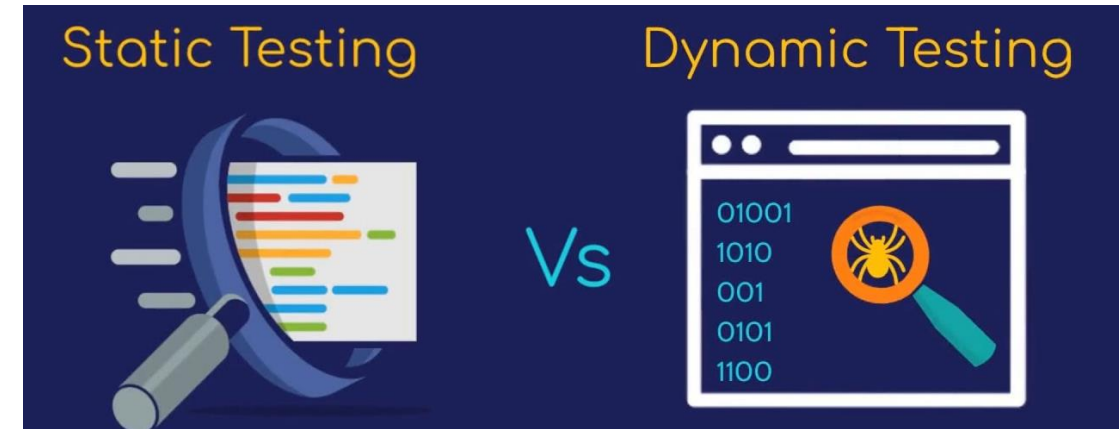
Types of software for testing - running

- **Static testing tools**

- testing without triggering
- code and syntax analysis
- manually / automatically

- **Dynamic testing tools**

- interaction with software
- providing information about the program from various scenarios



Types of software for testing - license

- **Open source tools**
 - freely available
 - customization option
 - e.g. Selenium , jMeter , appium
- **Tools sold**
 - need to buy a license
 - training fee but it is support
 - e.g. HP WinRunner , Silk Test, Ranorex
- **Created tool**
 - tailored to your needs
 - own development or supplied
 - e.g. Microsoft, IBM, Oracle...



closed source



open source

Most common software categories for testing

- agile testing tools
- automated testing tools
- mobile platform testing
- performance and load testing
- testing and management tools



Agile testing tools

- support for the development process
 - scrum , external programming, ...
- **Jira**
 - defect tracking
 - creating reports
 - project management
- **SoapUI**
 - support for service-oriented architecture and REST
- **Selenium**
 - highly widespread
 - support for multiple types of programming languages
 - JS, Node.js, Python , Ruby, Java, Kotlin , C # , Perl , Groovy



Automated testing tools

- running a large number of tests in a short time

- **HP UFT**

- commercial tool
- implementation of functional tests (GUI)
- mobile platforms and websites

- **Selenium**

- open source tool
- script automation + recording
- regression tests

- **Watir**

- open source tool
- Ruby library for testing web applications

- **Ranorex , IBM Rational Functional Tester, Tricentis Toca , Catalan Studio**



Mobile platform testing

- mobile test creation tools
 - automated UI tests
 - unit tests
 - mobile device test libraries
-
- **Appium Studio**
 - **Appium**
 - **SeeTest**
 - **eggPlant**
 - **Test Complete**
 - **Cobiton**



Performance and stress testing

- testing and verifying service performance
 - during operation and under load
- development of performance tests and stres

- **Apache Jmeter**
- **Tsung**
- **WAPT**
- **LoadNinja**
- **WebLOAD**
- **SmartMeter.io**



Testing and management tools

- use of web management tools
 - creating and running tests
 - recording tests
 - report generation
-
- **Zephyr**
 - **Qmeters**
 - **TestRail**
 - **TestLink**
 - **Qtest**
 - **TestLodge**



Tools for automated testing of web services

Web testing tools ⇅	Web browser based (model) ⇅	Scriptable ⇅	Scripting language ⇅	Recorder ⇅	Multiple domain ⇅	Frames ⇅
eggPlant Functional	Yes (IE, Firefox, Safari, Opera, Chrome)	Yes	SenseTalk	Yes		
iMacros	Yes (Firefox, Chrome, IE)	Yes	iMacro Script	Yes	Yes	Yes
Katalon Studio	Yes (IE, Firefox, Chrome, Safari, Opera and any modern browser)	Yes	Groovy	Yes	Yes	Yes
Ranorex Studio	Yes (Chrome, Firefox, Safari, IE)	Yes	C#, VB.NET	Yes	Yes	Yes
Sahi	Yes (IE, Firefox, Chrome, Safari, Opera and any modern browser)	Yes	Sahi Script	Yes	Yes	Yes
Selenium	Yes (IE, Firefox, Chrome, Safari, Opera, Edge)	Yes	Ruby, Java, NodeJS, PHP, Perl, Python, C#, Groovy	Firefox and Chrome	Yes	Yes
SOAtest	Yes	Yes	Python, JavaScript, Java	Yes	Yes	Yes
TestComplete	Yes (IE, Firefox, Chrome)	Yes	VBScript, JScript, C++Script, C#Script, DelphiScript, Python	Yes	Yes	Yes
Test Studio	Yes (Chrome, Firefox, Safari, IE)	Yes	C#, VB.NET	Yes	Yes	Yes
Tricentis Tosca ^[1]	Yes (Chrome, Firefox, IE, Edge)	Yes	C#, VB.Net	Yes	Yes	Yes
WatiN	Yes (IE, Firefox)	Yes	C#, ability to run JavaScript from C# calls	IE/FF		Yes
Watir	Yes (IE, Firefox, Chrome, Safari, Opera, Edge)	Yes	Ruby	No	Yes	Yes
HP-UFT	Yes (IE, Firefox, Chrome, Safari, Opera, Edge)	Yes	VBScript	Yes	Yes	Yes

https://en.wikipedia.org/wiki/List_of_web_testing_tools

We showed on Lab

- Selenium
- Robot Framework
- Postman



ROBOT
FRAME
WORK/



POSTMAN

An example of a " bug " that is also in the standard

- Excel
 - Lotus 1-2-3 compatibility
 - non-existent day 29.2.1900
- *A leap year occurs every four years. More precisely, it occurs in every year that is divisible by four, with the exception of round century dates (such as 1900, 2000, 2100), which are leap years only if they are divisible by 400 .*
- The bug is part of the international ISO and ECMA (European Computer Manufacturers Association),
 - bug standardized within the office Open XML document format



BUG.xlsx - Excel

Príhlásiť

Súbor Domov Vložiť Rozloženie strany Vzorce Údaje Revízia Zobraziť Pomocník Acrobat Povedzte, čo chcete urobiť Zdieľať

Calibri 11 A⁺

B I U

Všeobecné

Podmienené formátovanie Formátovať ako tabuľku Štýly buniek

Vložiť Odstrániť Formátovať

Zoradiť a filtrovať Najst' a vybrať Úpravy

Schránka Písmo Zarovnanie Číslo Štýly Bunky

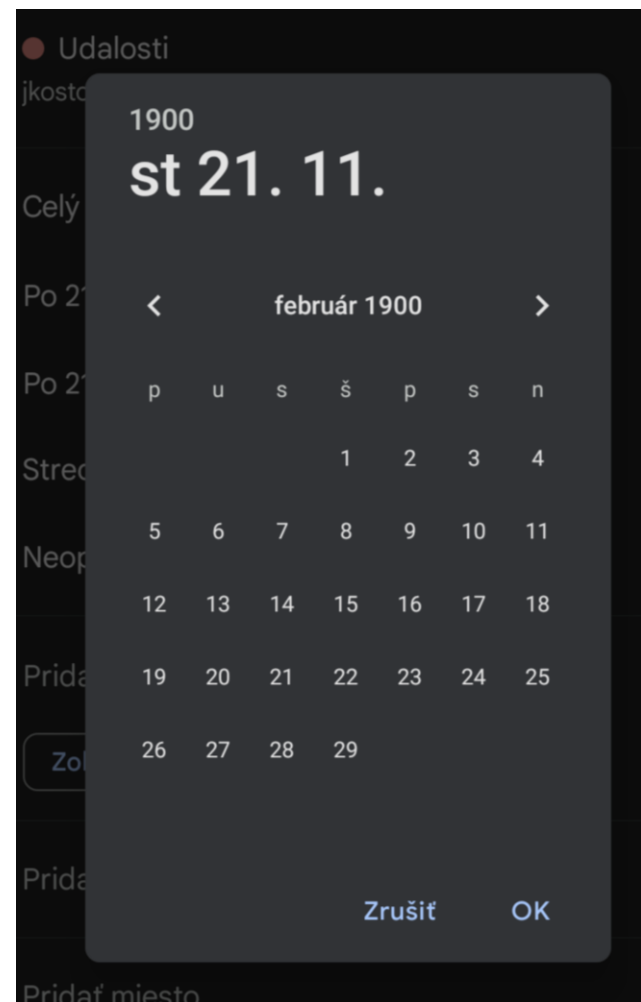
J9

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4		28.2.2016		28.2.1900	utorok	OK	28.2.1897	28.2.1897	
5		29.2.2016		29.2.1900	streda	OK	28.2.1898	28.2.1898	
6		1.3.2016		1.3.1900	štvrtok	OK	28.2.1899	28.2.1899	
7		2.3.2016		2.3.1900	piatok	X	29.2.1900	streda	
8		3.3.2016		3.3.1900	sobota	OK	28.2.1901	štvrtok	
9		4.3.2016		4.3.1900	nedeľa	OK	28.2.1902	piatok	
10		5.3.2016		5.3.1900	pondelok	OK	28.2.1903	sobota	
11		6.3.2016		6.3.1900	utorok	OK	29.2.1904	pondelok	
12									
13									
14									
15									
16									

Hárok1

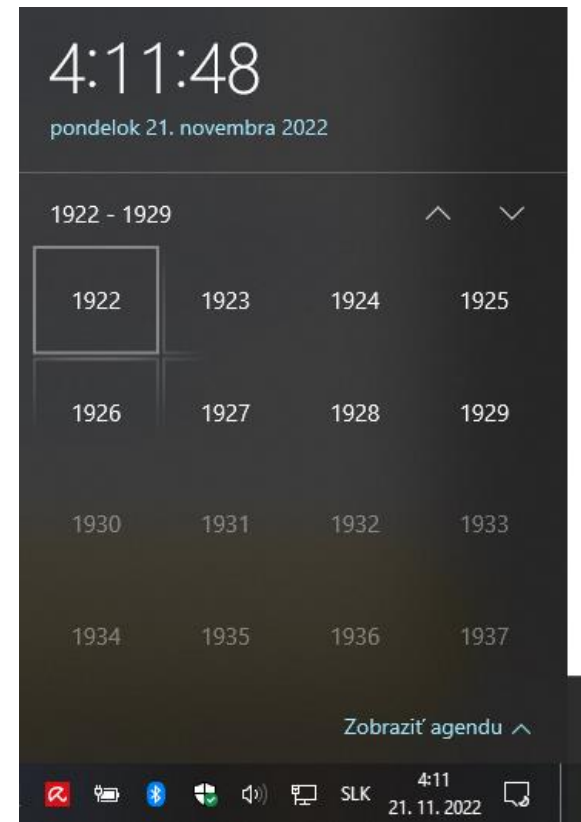
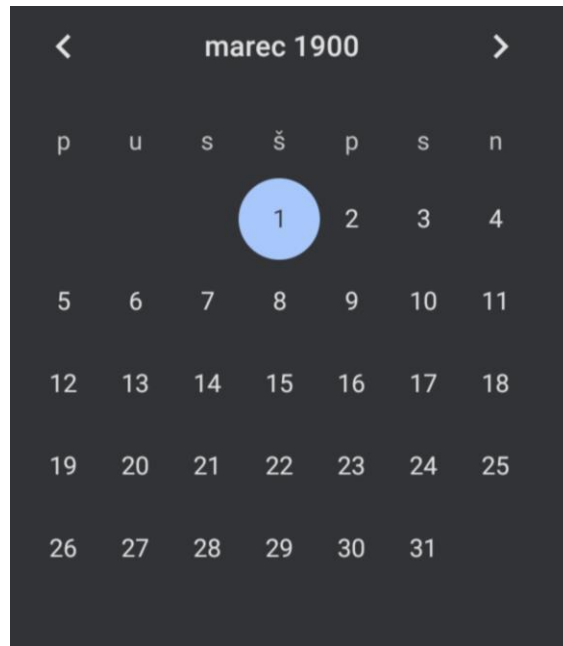
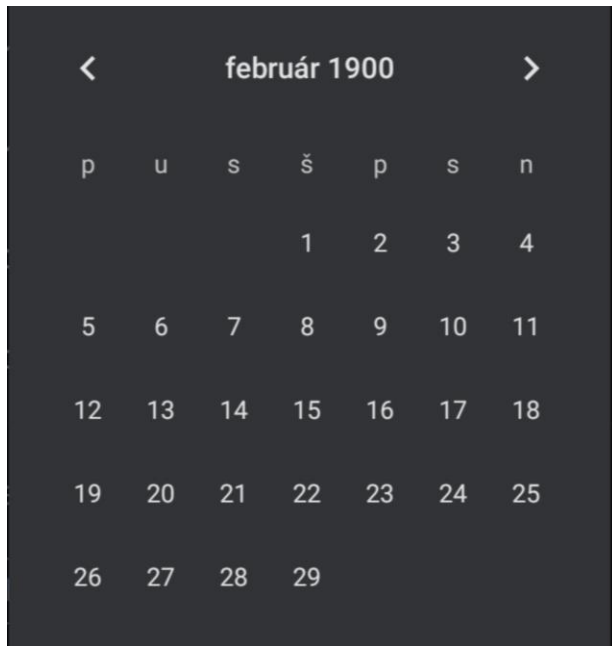
Prípravený

160 %



How to fix it?

- like Microsoft
 - do not display the given date
- Android displays the given day but corrects it when selected



Let's start the Software Testin



doc. Ing. **Jozef Kostolný**, PhD.
Fakulta riadenia a informatiky
Žilinská univerzita v Žiline
jozef.kostolny@fri.uniza.sk

Thank you for your attention